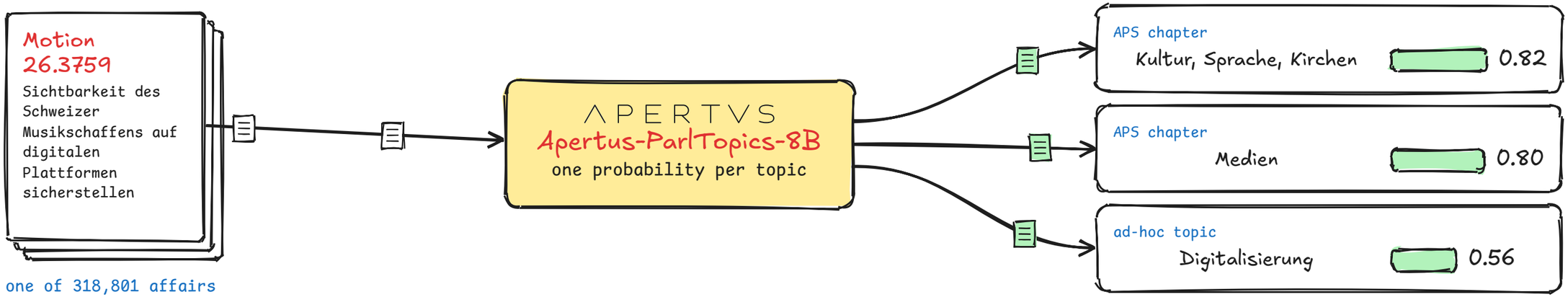
Parlamentsthemen

Topics for every Swiss parliamentary affair, in every national language, with Apertus

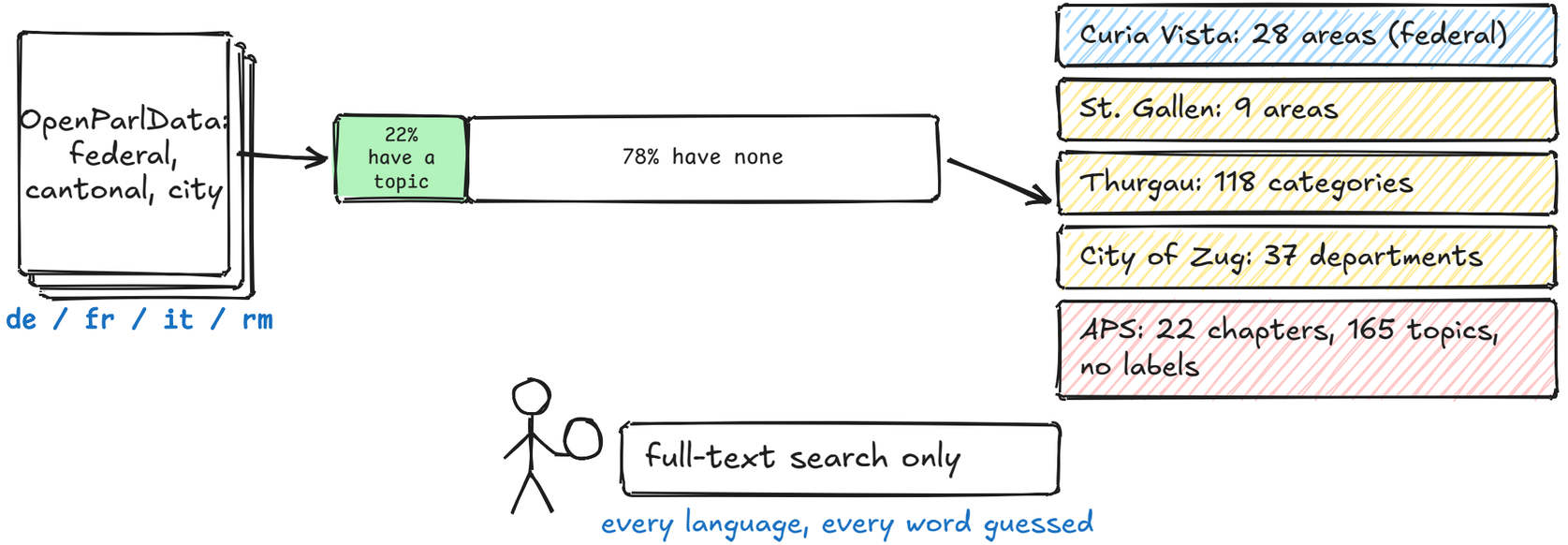


Invincibles

An open service that assigns standard and ad-hoc topics to the affairs in OpenParlData, shows the passage behind each suggested topic, and runs its trained Apertus model on a single GPU.

The Problem

Affairs from all Swiss parliaments are searchable mainly by full text, and topic labels exist only in pockets.[1]



318,801
affairs from 88 parliaments[2]

22%
carry a subject area from Curia Vista or a local scheme[3]

4 schemes
in four local parliaments, all different[3]

0
affairs labelled with Année Politique Suisse topics supplied for the challenge[4]

Digitale Verwaltung Schweiz
Administration numérique suisse
Amministrazione digitale Svizzera

Digital Public Services Switzerland (DVS) relies on PolitMonitor's manually assigned digitalisation tags[5] and needs such affairs found across the whole dataset.

1 Challenge background: https://gitlab.com/groups/opendata.ch/openparldatach/-/wikis/Home/Future-Topics/Classification
2 OpenParlData: https://openparldata.ch; label folder: https://files.openparldata.ch/hackapertus/ 3 Counts from our cleaned snapshot: Technical report, section 4
4 Année Politique Suisse topics: https://anneepolitique.swiss/de/themen/ 5 DVS list: https://www.digital-public-services-switzerland.ch/en/news/political-activities-regarding-digitalisation

A topic is a card

Interpretable by design

- One calibrated probability per topic, not a black-box label
- Topic cards in plain language: anyone can read and edit them
- Every suggestion shows the passage and the reason behind it

Motion 26.3759, 18.06.2026: Sichtbarkeit des Schweizer Musikschaffens auf digitalen Plattformen sicherstellen

Questions: Digitalisierung and 28 Curia Vista areas; APS and ad-hoc topics work the same way. Assigned by the Parliamentary Services: Wirtschaft, Medien und Kommunikation, Kultur.

Chat model

Apertus 8B, prompted for a JSON answer — tokens written: 215 (one token after another)

```
{
"Digitalisierung": true,
"Wirtschaft": true,
"Gesundheit": false,
"Staatspolitik": true,
"Finanzwesen": false,
"Umwelt": false,
"Internationale Politik": false,
"Soziale Fragen": false,
"Sicherheitspolitik": false,
"Verkehr": false,
"Migration": false,
"Medien und Kommunikation": true,
"Landwirtschaft": false,
"Energie": false,
"Beschäftigung und Arbeit": false,
"Europapolitik": false,
"Recht Allgemein": false,
"Bildung": false,
"Sozialer Schutz": false,
"Raumplanung und Wohnungswesen": false,
"Steuer": false,
"Menschenrechte": false,
"Strafrecht": false,
"Kultur": true,
"Wissenschaft und Forschung": false,
"Parlament": true,
"Zivilrecht": false,
"Internationales Recht": false,
"Gerichtswesen": false
}
```

A string to parse: true or false, no probability. It also marks Staatspolitik and Parlament, which the Parliamentary Services did not assign.

Decision model

Apertus 8B with a trained adapter — tokens written: 0 (one forward pass)

Digitalisierung	0.56	Europapolitik	0.00
Wirtschaft	0.38	Recht Allgemein	0.01
Gesundheit	0.00	Bildung	0.00
Staatspolitik	0.04	Sozialer Schutz	0.00
Finanzwesen	0.02	Raumplanung und Wohnungswesen	0.00
Umwelt	0.01	Steuer	0.01
Internationale Politik	0.02	Menschenrechte	0.00
Soziale Fragen	0.03	Strafrecht	0.00
Sicherheitspolitik	0.01	Kultur	0.89
Verkehr	0.00	Wissenschaft und Forschung	0.00
Migration	0.00	Parlament	0.00
Medien und Kommunikation	0.88	Zivilrecht	0.00
Landwirtschaft	0.00	Internationales Recht	0.00
Energie	0.00	Gerichtswesen	0.00
Beschäftigung und Arbeit	0.00		

One probability per question. A topic is marked when its bar passes the blue tick, a threshold fixed on earlier affairs. Here the marks match the assigned topics.

1 215 tokens

The chat model writes token by token: 215 to 244 tokens and 1.4 to 2.4 s per affair (5 runs, hosted endpoint).

2 0 tokens

The decision model writes nothing. One forward pass gives a probability for each question.

3 1.2 s per affair

28 questions per pass, timed over 1,000 test affairs on one RTX A6000: about 3,000 affairs per GPU hour.

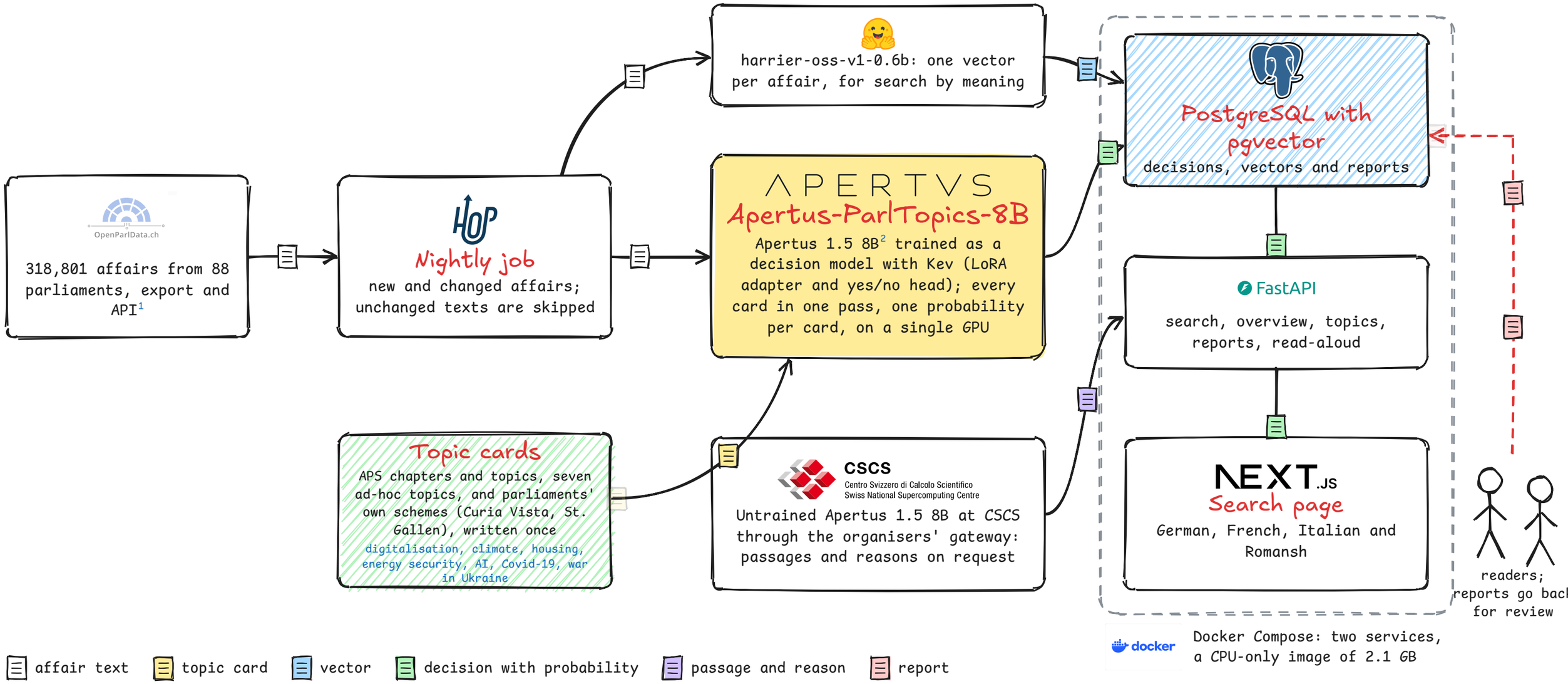
4 AP 0.74 · F1 0.74

APS chapters: macro AP 0.74 on 1,000 federal affairs. Seven ad-hoc topics: F1 0.74 on 400 affairs.

How we trained it: a decision model[1] built with Kev[2] on Apertus 1.5 8B[3]. 6,728 federal affairs from before 2024, with yes/no questions from Curia Vista labels (23 subject areas, 400 descriptors); a rank-16 LoRA adapter and a yes/no head on unchanged base weights; one epoch, 1 h 48 min on one A6000; per-topic Platt calibration and thresholds. APS, cantonal schemes and digitalisation: no labels used in training.

1 TypeSafe, "Introducing System One models and Jev": https://typesafe.ai/blog/introducing-system-one-models-and-jev 2 Kev: https://github.com/jaredpalmer/kev Model card: https://huggingface.co/swiss-ai/Apertus-v1.5-8B
3 Project Apertus (ETH Zurich, EPFL, CSCS), "From Open Text Models to Open Multimodal Reasoners: Apertus v1.5 Technical Report", October 2026: https://huggingface.co/papers/2610.15542

How the Service Runs



1 OpenParlData: https://openparldata.ch 2 Apertus 1.5 8B model card: https://huggingface.co/swiss-ai/Apertus-v1.5-8B

Results, Coverage and Comparison

APS, two levels[1]

No APS labels used in training

0.74
APS chapters, 1,000 federal affairs
ranking quality (AP); random: 0.15

0.66 / 0.57
APS chapters, cantonal / city affairs
agreement (F1) with Claude Opus 5.5[2]

0.61
APS topics, second level
agreement (F1) with Claude Opus 5.5

AP: 1.0 means every right affair ranked first. F1: 1.0 means perfect agreement.

Four national languages[8]

Same quality in every language: German 0.71 · French 0.70 · Italian 0.70
Curia Vista macro AP on the same 871 affairs

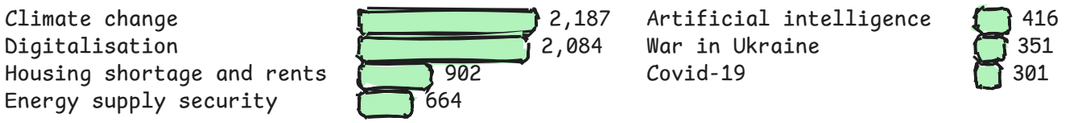
A query in one language finds affairs in all others
Interface, search and read-aloud also in Romansh

Ad-hoc topics, no retraining

0.74
7 ad-hoc topics, 400 affairs
agreement (F1) with Claude Opus 5.5; digitalisation 0.84, climate 0.67

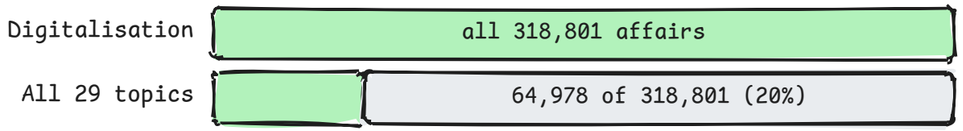
29 of 30
DVS-listed digitalisation affairs found[3]
weekly monitoring shortlist

0.70
30 new topics, each from one keyword
ranking quality (AP); random: 0.02



Affairs labelled per ad-hoc topic, 2024 to 2026 (49,978 affairs); each topic is one card, added without retraining.

Stored decisions today



All 29 topics: 2024 to 2026 and the first 15,000 from 2023, newest first.

Backfill: about 8,000 affairs per hour on three GPUs; the remaining 254,000 affairs need about 90 GPU-hours (about 30 hours on three GPUs). New affairs: about one GPU-minute per night.

As good as a frontier model[6]

Micro F1, full test sets, official labels	Claude Opus 5.5	Apertus-ParlTopics-8B
St. Gallen, 868 cantonal affairs	0.70	0.71
Curia Vista, 1,000 federal affairs	0.69	0.70
APS chapters, 1,000 federal affairs	0.62	0.63

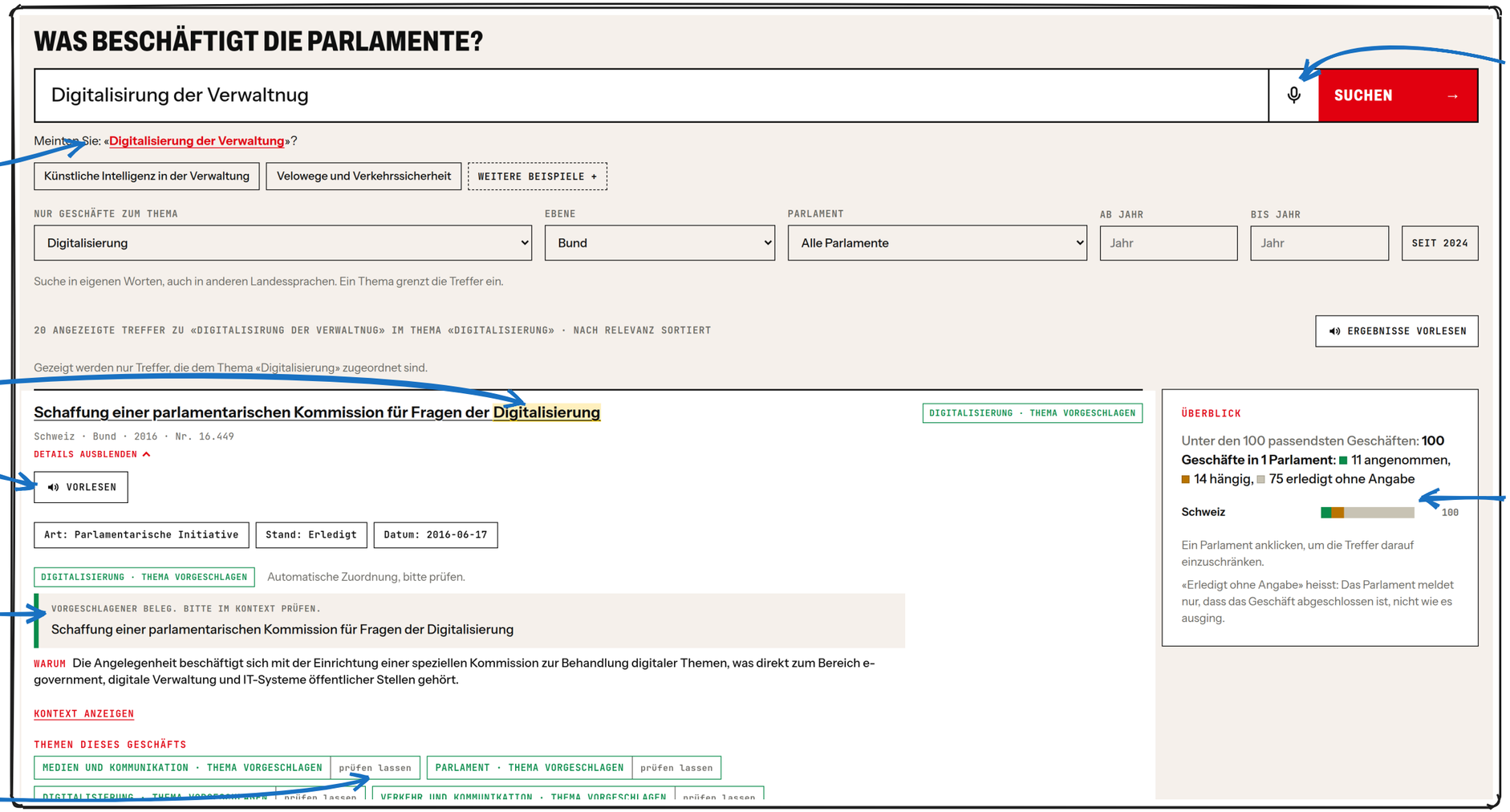
Claude Opus 5.5: closed frontier model, size not disclosed, external API. Apertus-ParlTopics-8B: open, 8 billion parameters, runs on a single GPU. Differences within the 95% interval: level, ours ahead on every point estimate.

System	Trained on	Digitalisation AP	Curia Vista AP (28 areas)	New scheme or topics, no labels (macro AP)[7] St. Gallen (new scheme)	30 new topics
Legal XLM-RoBERTa (2023 study)[2]	20,000 labelled affairs, only these 28 areas, 3 epochs	0.42	0.73	no output	no output
Embeddings and classifier (Demokratis)[4]	20,000 labelled affairs, only these 28 areas	0.47	0.74	0.38	0.39
Apertus-ParlTopics-8B (ours)	6,728 labelled affairs, 1 epoch; one model for every scheme	0.46	0.71	0.70	0.70
Apertus-ParlTopics-8B plus embedding classifier	as above	0.49 (best of 23)	0.76 (best)[5]		

Combined with an embedding classifier, Apertus-ParlTopics-8B is the best system on both digitalisation (0.49) and Curia Vista (0.76); on its own it also handles new topics and schemes without new training.
Ahead of untrained Apertus 8B and 70B, open decision models (Kev-9B, decisio) and the 2023 legal XLM-RoBERTa on digitalisation; handles new topics and schemes without new training.

1 APS topics: https://anneepolitique.swiss/de/themen/ 2 Our evaluation, the baselines and the frontier-model judge (Claude Opus 5.5 and Claude Sonnet 5.5): Technical report, section 4
3 DVS list: https://www.digital-public-services-switzerland.ch/en/news/political-activities-regarding-digitalisation 4 Demokratis: https://github.com/Demokratis-ch/demokratis-ml
5 Comparison and blend (a logistic blend per area, fitted on earlier affairs): scripts/evaluation/metrics_build.py 6 Full frontier comparison: scripts/evaluation/judge_extend.py 8 Technical report, section 4
7 New-scheme scores: scripts/evaluation/zero_shot_embeddings.py (embeddings matched to the topic card text, no labels); the legal XLM-RoBERTa has no output there: it can only score the 28 areas it was trained on

Accessible to Readers, Efficient to Run



"Did you mean": corrects typos

Speak your search

DE FR IT RM EN

5 languages

Search words marked

Read aloud[1] IIElevenLabs

Outcomes per parliament

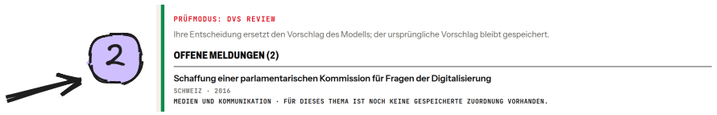
Passage and reason behind each topic

Keyboard and screen reader ready

People correct topics



Reader clicks "prüfen lassen" on a doubtful topic



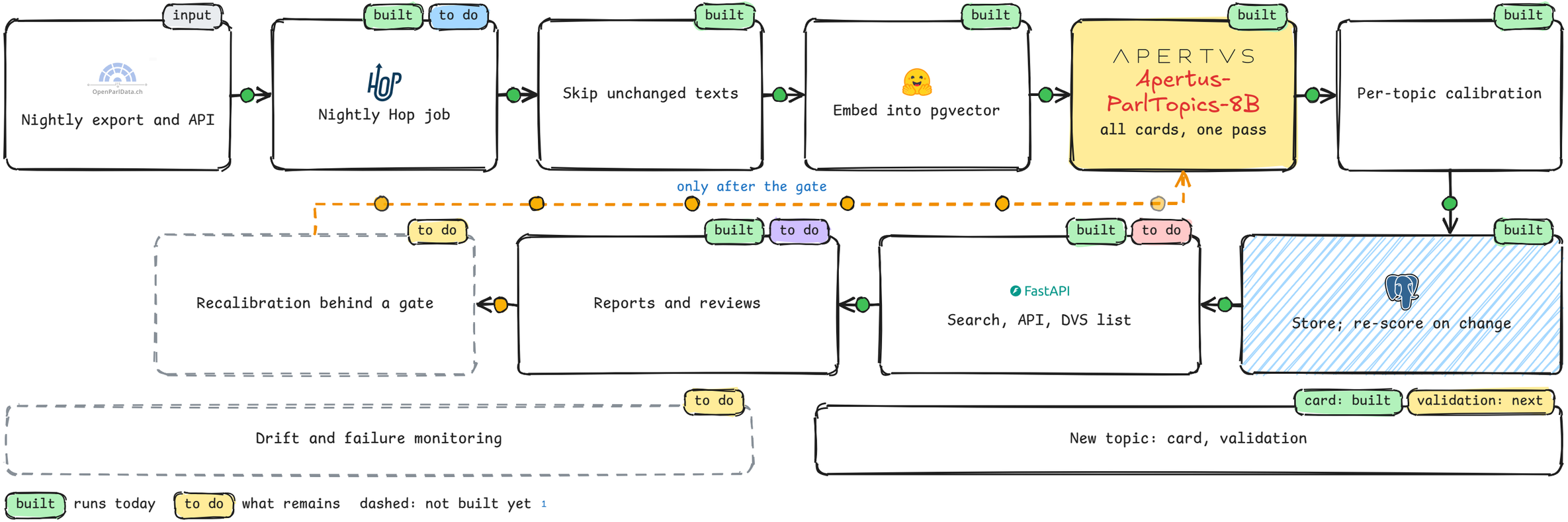
Report lands in the reviewers' queue



Reviewer confirms, removes or adds; everyone sees ✓ geprüft, the model's original kept

1 Read-aloud voices: ElevenLabs models, https://elevenlabs.io/docs/overview/models

The Pipeline: Built and What Remains



Value

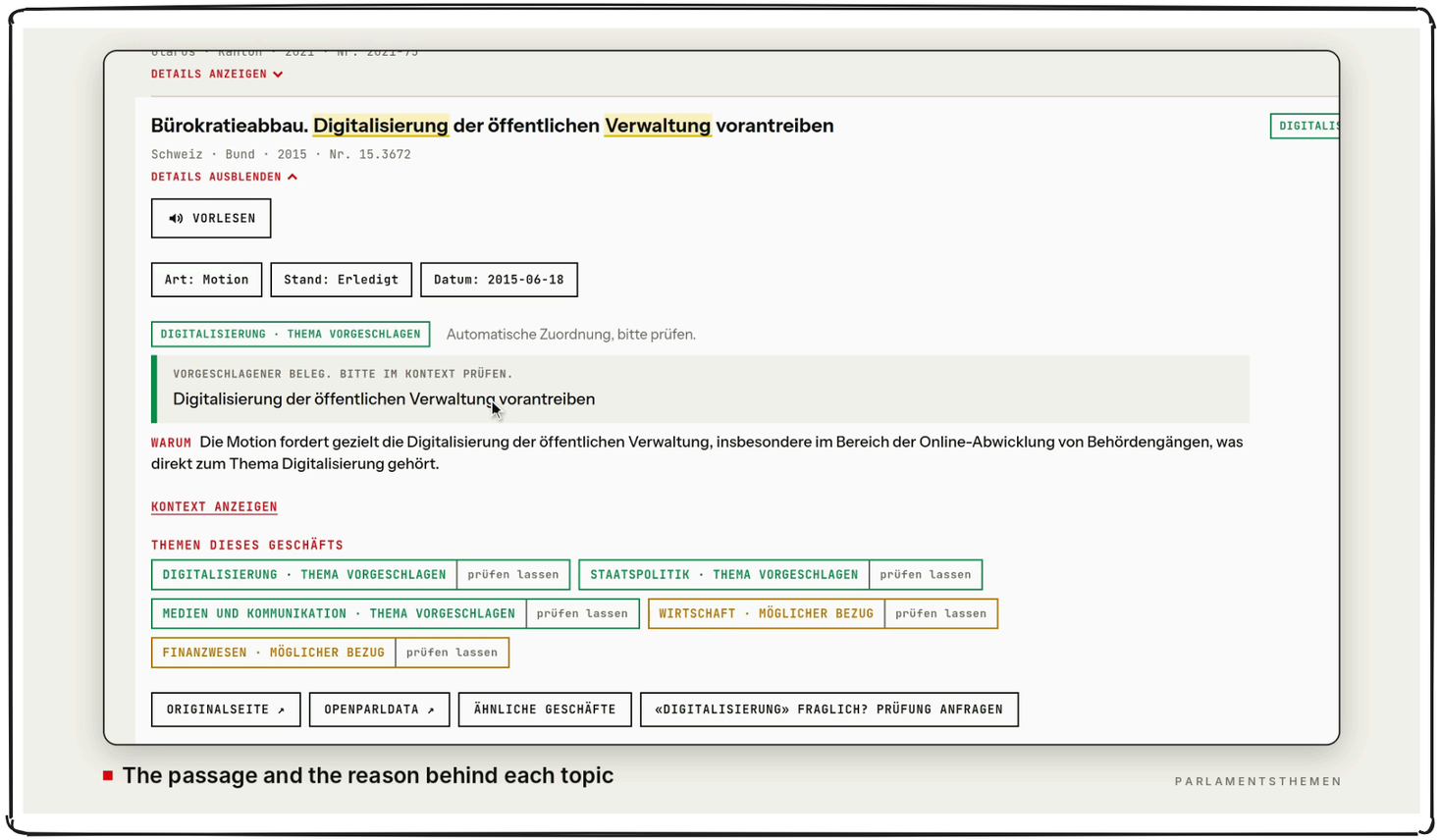
- 29 of 30 DVS affairs on the weekly list[2]
- Open models, a single GPU
- About one GPU-minute per night

What remains

Schedule the nightly job; finish the backfill → Review with DVS staff → Recalibration gate; drift alerts → DVS pilot: weekly shortlist

1 Pipeline status: Technical report, section 2 2 DVS list: https://www.digital-public-services-switzerland.ch/en/news/political-activities-regarding-digitalisation

Live Demo, Team and Thanks



Parlamentsthemen · Invincibles

Model: huggingface.co/merabro/Apertus-ParlTopics-8B

Code: github.com/prakash-aryan/parlamentsthemen

Invincibles



Kaushik Raghupathruni



Vilas Harapanahalli



Vaibhav Motwani



Prakash Aryan

With thanks to

swissAI Swiss AI Initiative

CSCS Centro Svizzero di Calcolo Scientifico Swiss National Supercomputing Centre

ElevenLabs

OpenParlData.ch

Hack Apertus

Digitale Verwaltung Schweiz Administration numérique suisse Amministrazione digitale Svizzera

DVS

Open source: Kev, harrier-oss, FastAPI, PostgreSQL with pgvector, Next.js and Apache Hop.